

Online Appendix to “The Effect of the Voting Rights Act on Enfranchisement: Evidence from North Carolina”

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Note that this Appendix is published online only.

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A Data Sources, Measurement and Descriptives

This appendix presents additional information about the data used in paper’s empirical analysis, including analysis that is specific to the appendices. Specifically, I provide a table that collects all of the data sources and their coverage of counties and years. I next describe in more detail the measurement of the voter registration data with particular attention to the denominator in the variable (registered as a percentage of *the relevant voting age population*). Finally, I present basic descriptive statistics of the variables for the entirety of the sample period, and additionally for the pre-VRA period subsetting for covered and uncovered counties separately.

Tables A2 and A3 collect information on all of the data sources used in this paper. Given the importance of data transparency to reproduce and build on existing findings, clarity in data collection is of significant importance. Moreover, the pre-VRA voter registration data by race constitutes an important improvement on data previously used in the literature. The availability of two years of pre-VRA data allows for an estimation of pre-VRA trends that are essential to help substantiate the assumption that trends in political outcomes would have continued the same in covered counties had Section 5 not been applied (see Online Appendix C). The VRA constituted the impetus for significant data collection efforts undertaken at the local level with respect to racial and political variables in the US. As a result, data on pre-VRA enfranchisement at the local level in the South is generally quite limited.

Table A1: Summary Statistics

Variable	Mean	SD	Min	Max	N
Black voter registration	0.68	0.31	0.00	3.03	1695
White voter registration	0.78	0.19	0.06	2.13	1694
White-black registration gap	0.10	0.36	-1.68	2.00	1694
Voter turnout, presidential	0.50	0.14	0.17	0.93	1200
Voter turnout, congressional	0.42	0.21	0.04	0.96	1100
Democratic vote share, presidential	0.50	0.16	0.14	0.96	1200
Democratic vote share, governor	0.63	0.16	0.06	0.98	800
ln(population)	10.51	0.95	8.24	13.45	1100
% population black	0.24	0.17	0.00	0.66	588
Median household income	13940	12287	750	54988	600
Median years of schooling	9.50	1.66	6.90	14.10	400

Notes: Summary statistics in The table are presented for all counties in all years for which data for the variable is available. Temporal data availability differs by measure. See Table A3. I present un-interpolated summary statistics for variables that are interpolated in the paper. I exclude variables — lynchings and incarceration rates — that are not included in any of the main specifications of the table and are only measured in the pre-VRA period. Table A4 includes these variables as it examines only pre-1965 levels. An observation (N) is a county-year pair. The text describes the missing voter registration data. There are also missing years of black population data as a result of suppression of small minority populations from the census data. This is for one year only — 1970 — and I am able to interpolate for these missing observations using 1960 and 1980 data.

The main analysis of the paper utilizes voter registration rates by race as a percentage of the relevant voting age population. Registration before 1970 is calculated as a percentage of the voting age population by some of the sources of registration data used (see Table A3). There

is likely to be measurement error in the sources' calculation of the denominator, as the sources only had access to the previous decennial census population data (and potential projections of population). In the main analysis of the paper I use registration *rates* from the sources when available, and calculate rates using total registrants when rates are unavailable.¹ In Appendix H, I re-calculate the registration percentages using aggregate registration data where available (all years except 1958) for the numerator, and linear interpolated census data to calculate the relevant denominators. The results are similar, but smaller in magnitude.

To construct the percentages for the post-1970 registration data, it's necessary to use census data to measure the denominator (*relevant voting age population*). The minimum voting age in North Carolina was 21 years before 1971, and 18 years afterwards. Measuring the relevant voting age population becomes challenging because the racial categories *black* and *non-white* in the voter registration data do not perfectly correspond to the population data within and across years for North Carolina. Since non-black minorities were a small percentage of the North Carolina population during the time period of interest, the paper assumes that the measurement error introduced by the discrepancies in these racial categories is small. In 1960, for instance, blacks accounted for 96.5% of non-whites in North Carolina (United States Commission on Civil Rights, 1961). The following years have population-by-age-by-race and voter registration data that do not match on racial categories: 1970, 72, 74 and 76. For each of those years, voting age population is calculated the *black* category while aggregate voter registration data is for the *non-white* category. For all other years, registration and age data are either both for the *black* category or both for the non-white category. Wherever possible, I utilize the *black* category.

Registration data by race is not available for all counties in all years.² However, even with the missing observations, the panel is highly balanced — 13 of 1,500 non-white registration observations are missing, and 6 of 1,500 white observations. Moreover, the counties that are missing data are *uncovered* counties, thus assuaging the concern that covered counties failed to report registration statistics because of discriminatory practices that might upwardly bias this paper's results for black/non-white registration. I make the following corrections to account for the missing observations. Graham county is missing data for 1966, 1971, and biennially from 1972-1984. However, Graham had a black population in single digits in these years. Moreover, Graham *never* had a black registered voter in any year when registration data is available. I therefore assign black voter registration of zero to the missing years (and thus 0% registered) based on this information. While it is not known, this strategy is based on the assumption that missing observations for Graham are the result of data suppression to ensure confidentiality given the small black population. Finally, results dropping Graham entirely, dropping 1966 and 1967 (most data is missing for these years),

¹I make the following correction to total registrations in 1958. Swain county's non-white registrants are calculated as negative in 1958 based on the difference in total registrants provided by the source. See Table A2. To correct this negative value, I use source's measure of the percentage point change in registration assuming a constant denominator from the source's 1960 denominator.

²Graham county is missing black/non-white data in 1966, 1971, and biennially from 1972-1984, and white registration data in 1966. Dare county is missing all data by race from 1966. Davidson and Orange counties are missing all data by race for 1967. Yadkin county is missing all data by race for both 1966 and 1967.

and dropping the remaining counties that are missing data are similar to the main results of the paper and available upon request.³

The differences-in-differences approach used in the main analysis of the paper employs fixed effects to account for time non-varying differences between covered and uncovered counties that might affect both coverage status and subsequent political outcomes. However, readers may be interested in the descriptive difference in levels for various variables in the pre-VRA period. I provide this descriptive information in Table A4. The table includes two variables — lynching victims and the black incarceration rate — that are only available in the pre-VRA period. Many political outcomes indicate that eventually-covered counties were more discriminatory to minorities (e.g. black voter registration rates are lower in counties that were eventually covered), though there were slightly more acts of racial violence in counties that remained uncovered post-VRA.

Table A2: Data Sources for Voter Registration

Year	Measurement & Source
1958	Percentage point change in % registered (and absolute change in number registered) between 1958 and 1960 provided by the source; percentages of the relevant voting age population for white/non-white in 1958 calculated based on those changes (United States Commission on Civil Rights, 1961, 499-501).
1960	Percentages of the relevant voting age population (and totals) for white/non-white provided by the source(s) (United States Commission on Civil Rights, 1961, 499-501) and (United States Commission on Civil Rights, 1962, 26-28).
1966	Percentages of the relevant voting age population (and totals) for white/black provided by the source as of summer 1966 (Voter Education Project, 1966).
1967	Percentages of the relevant voting age population (and totals) for white/non-white provided by the source as of February 1967 (United States Commission on Civil Rights, 1968, 248-251).
1968	Percentages of the relevant voting age population (and totals) for white/black provided by the source as of summer 1968 (Voter Education Project, 1968).
1970-1988	Voter registration totals for white/black available every two years provided by the source (ICPSR, 1991).
1971	Voter registration totals for white/non-white provided by the source (Brock, 1972). Data available for 99 counties.
1993	Voter registration totals for white/black provided by the source (North Carolina State Board of Elections, 1993).

³Of course dropping the observations is not the best approach to unbalanced panel analysis, but given the scarcity of missing observations and the complexity of unbalanced panel corrections, this seems the most reasonable approach.

Table A3: Data Sources

Variable	Measurement & Source
Covered	Equal to 1 for counties covered by the 1965 version of the act and equal to 0 otherwise. Jackson county was covered in the 1970 extension of the act and Wake was bailed out of coverage in 1984. Wake was covered for the majority of years of study and so is coded as covered. Jackson's coverage indicator begins in 1970 (United States, 1965, 1966 <i>b,c</i>).
Voter turnout	
Congressional, 1952-1972	Total votes cast for President provided by the source (ICPSR, 1995). Measured in the paper as a percentage of the total voting age population.
Presidential, 1948-1992	Total votes cast for Congressional Representative provided by the source (Leip, 2012). Measured in the paper as a percentage of the total voting age population.
Democratic two-party vote share	
Presidential, 1948-1992	Percentage of total votes for President cast for the Democratic party candidate provided by the source (Leip, 2012).
Gubernatorial, 1952-1984	Percentage of two-party votes for governor cast for the Democratic party candidate provided by the source (State of North Carolina, 1972, 1986).
Voting age population by race, 1950-2000	Voting age population calculated as 20+ (as 21+ is not available) before 1971 and 18+ after 1971 for decennial census years; racial categories are black or white (Minnesota Population Center, 2011). Non-recorded years linearly interpolated. See text for more information on pre-1971 voting age.
Population by race, 1950-90	Source provides decennial census years; racial categories are black or white (Minnesota Population Center, 2011). Non-census years linearly interpolated.
Median household income, 1949-1989	Source provides the year prior to decennial census years; income measured in current dollars (Minnesota Population Center, 2011). Non-census years linearly interpolated.
Median years schooling, 1950-1980	Source provides decennial census years; non-census years linearly interpolated; measured for the population aged 25+ (Minnesota Population Center, 2011).
Lynching victims, 1883-1930	Cumulative (total) lynching victims over the period (Hines and Steelwater, 2015). No victims are recorded by source post-1930.
Black incarceration rate, 1960, 1962	Black state prisoner and total state prisoner intake provided by source (North Carolina Prison Department, 1962, 1964). Black intake calculated as percent of total intake.

Table A4: Summary Statistics by Coverage Status, Pre-1965 Levels

Variable	Covered				Uncovered			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Black voter reg.	0.231	0.168	0.054	1.305	0.560	0.448	0.000	3.030
White voter reg.	0.862	0.242	0.379	1.521	1.077	0.267	0.482	2.129
White-black reg. gap	0.631	0.313	-0.571	1.317	0.516	0.484	-1.678	1.918
Voter turnout, pres.	0.371	0.090	0.168	0.570	0.609	0.158	0.225	0.930
Voter turnout, cong.	0.261	0.135	0.042	0.563	0.541	0.211	0.077	0.962
Dem. vote share, pres.	0.716	0.114	0.401	0.965	0.516	0.127	0.195	0.880
Dem. vote share, gov.	0.794	0.119	0.420	0.978	0.573	0.152	0.061	0.954
ln(population)	10.494	0.786	8.561	12.415	10.193	0.880	8.416	12.514
% population black	0.400	0.129	0.127	0.642	0.168	0.150	0.000	0.664
Median hh income	2423	967	750	5417	2609	1183	750	5632
Median yrs schooling	8.156	0.916	7.100	11.600	8.241	0.854	6.900	11.600
Lynching victims	1.880	0.994	1.000	4.000	2.040	1.484	1.000	6.000
Black incarceration rate	0.596	0.231	0.120	1.000	0.491	0.266	0.031	1.000
Counties	40				60			

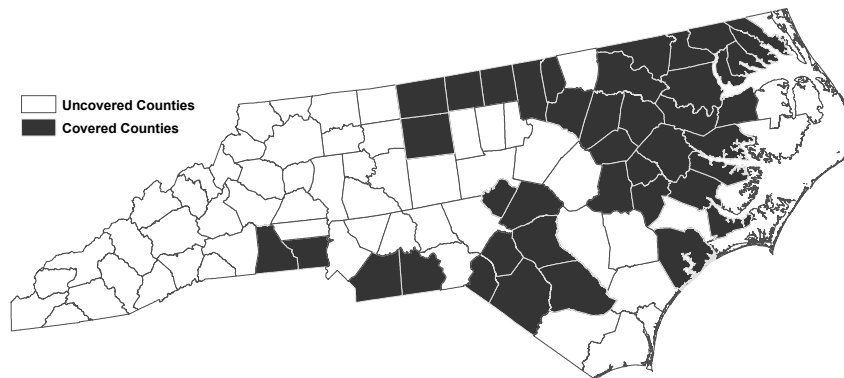
Notes: Summary statistics in the above table are presented for all counties in pre-1965 years for which data for the variable is available. The number of years varies by measure. See Table A3 for more details. Note that these level differences in pre-VRA outcomes are accounted for in the estimation procedure by including county fixed effects.

B Section 5 Coverage in North Carolina

The research design utilized in the paper’s empirical analysis relies on an (arguably) arbitrary political participation threshold applied in 1965. In particular, 40 of North Carolina’s 100 counties were covered by Section 5 of the VRA because they failed to turnout at least 50% of their voting age population in the 1964 presidential election. Figure B1 shows the location of these counties. Given the importance of this differential application of Section 5 coverage, this section provides more detail about Section 5 and the application of the political participation threshold.

Figure B1: North Carolina Counties Covered by Section 5

Counties in North Carolina covered by the VRA were clustered in the state’s Black Belt.



Notes: Counties in North Carolina covered by Section 5 of the VRA are shaded, while un-covered counties are white. See Online Appendix B for a list of the counties and their voter turnout rates in 1964 that determined coverage status.

The VRA was signed into law in August of 1965 by President Lyndon Johnson on the heels of the three previous civil rights acts of 1957, 1960 and 1964. Many southern states, while abiding by the letter of the three pre-existing laws, had nevertheless taken to using less explicit forms of discrimination — literacy tests, poll taxes and grandfather clauses, among others — to keep blacks off the registration rolls and away from the polls. The VRA abolished these methods of disenfranchisement, succeeding where previous laws had failed by establishing provisions for enforcement.⁴ The primary mechanism of enforcement was Section 5, which required covered jurisdictions to submit all changes to their voting rules to the Department of Justice for examination prior to those changes going into effect, a practice known as *pre-clearance*. By requiring pre-clearance, the VRA curbed the cat-and-mouse game between the Federal government and the southern states, which had evolved as states attempted to devise new methods of keeping blacks out of politics, and the Federal government tried to keep up in rendering those methods illegal.⁵

⁴In particular, Section 2 of the VRA abolished “voting qualifications or prerequisites to voting” that denied the right to vote on the basis of “race or color.” Substantively, however, Section 2 merely reiterated provisions of existing law.

⁵As Governor James Coleman of Mississippi, a segregationist and opponent of federal civil rights legislation, put it: “Any [state] legislature can pass a law faster than the Supreme Court can erase it” (Bullock and Gaddie, 2009, 11). In upholding Section 5 of the VRA in place of case-by-case ex-post litigation of discriminatory practices,

The 1965 VRA was applied to most southern states in their entirety.⁶ Counties in North Carolina, however, were selectively covered by Section 5 based on a triggering formula established with regard to pre-1965 political participation and applied universally to all states. Of North Carolina’s 100 counties, 40 were covered by Section 5, while the remainder were left uncovered.⁷ The language of the triggering formula under Section 4(b) of the 1965 VRA states that Section 5 coverage:

...shall apply in any State or in any political subdivision of a state which (1) the Attorney General determines maintained on November 1, 1964, any test or device, and with respect to which (2) the Director of the Census determines that less than 50 per centum of the persons of voting age residing therein were registered on November 1, 1964, or that less than 50 per centum of such persons voted in the presidential election of November 1964.

That is, Section 5 applied to those states or political jurisdictions that maintained a discriminatory voting practice — a “test or device” — *and* that did not meet a political participation threshold — *either* in terms of registration or turnout — established at 50%. North Carolina maintained a literacy test in 1964 and thus satisfied the first requirement for coverage.⁸ Documentation from the Federal Register indicates that the latter threshold — 1964 voter turnout — was applied in North Carolina.⁹ This application of coverage to North Carolina counties based on this 50% threshold for voter turnout in a pre-VRA presidential election (1964) constitutes the assignment to treatment that this paper exploits to estimate the geographic counterfactual, and thus the causal effect of Section 5 coverage.

Table B1 shows voter turnout as a percentage of the voting age population and coverage status for North Carolina counties. This paper does not have access to the specific 1964 voter turnout calculations that the Bureau of the Census used to determine coverage. To the best of the author’s knowledge, no publication, government or otherwise, has reproduced the precise measurement of the (forcing, or intent-to-treat) variable that was ultimately used in the Department of Justice’s

the US Supreme Court stated: “...case-by-case litigation was inadequate... because of the inordinate amount of time and energy required to overcome the obstructionist tactics invariably encountered in these lawsuits. After enduring nearly a century of systematic resistance to the Fifteenth Amendment, Congress might well decide to shift the advantage of time and inertia from the perpetrators of the evil to its victims” (US Supreme Court, 1966).

⁶Alabama, Georgia, Louisiana, Mississippi, South Carolina and Virginia were covered in their entirety. When the pre-clearance statute was extended in 1970 and 1975, additional state and sub-state jurisdictions were added — Alaska, Arizona and Texas, along with counties in California, New York, Florida, and South Dakota, and municipalities in New Hampshire and Michigan.

⁷It’s important to note that this triggering formula was not designed for North Carolina specifically. It was applied universally to all states. One county — Jackson — was covered by the 1970 extension of Section 5 of the VRA, while one county — Wake — “bailed out” (was relieved) of its pre-clearance requirement in 1984. Jackson is considered covered post-1970 for the purposes of the paper’s analysis. Wake is considered covered for the entirety of the period to avoid bias from selection out of coverage. Moreover, Wake was covered for the majority of post-treatment years for which outcome data is available.

⁸Article VI, Section 4, of the North Carolina state constitution required a literacy test of all potential voter registrants (United States Commission on Civil Rights, 1962). The section read: “Every person presenting himself for registration shall be able to read and write any section of the Constitution in the English language.”

⁹See United States, 1965, 1966*b,c*. Online Appendix B provides more details about the determination of the threshold, its measurement and its application in North Carolina.

application of coverage.¹⁰ This paper has, however, verified total county-level voter turnout for North Carolina in the 1964 presidential election from a number of sources (North Carolina State Board of Elections, 1965; Leip, 2012; ICPSR, 1995).

Estimates of the voting age population, however, differ by source. None of the sources of voting age population consulted by this paper, when combined with the verified total county-level turnout numbers to construct a variable for 1964 voter turnout, result in the perfect prediction of how coverage is known to have been applied in North Carolina. Table B2 details non-compliance by these different measures of the voting age population. From the perspective of the difference-in-differences estimation, this non-compliance is not problematic as the estimation leverages variation in actual treatment status, with intent-to-treat comprising part of the county fixed effects. Nevertheless, it's worth noting the likely existence of non-compliers. In the local average estimates presented in Online Appendix D, the VA measure is used to determine which counties were "close" to the threshold for coverage. I use the VA measure because it produces the fewest number of non-compliers, and because it's possible that the US Civil Rights Commission had access to more accurate data on the voting age population than interpolated census data can provide.¹¹

¹⁰A FOIA request submitted in July 2012 to the Department of Justice, the department which received the Census Bureau's coverage determinations, did not return any documentation on these voter turnout calculations.

¹¹Or alternatively, even it was less accurate, that it was closer to what was actually utilized to apply coverage.

Table B1: 1964 Voter Turnout and Section 5 Coverage in North Carolina by County

County	1964 Voter Turnout	Covered	County	1964 Voter Turnout	Covered
Alamance	59.2%		Franklin	43.0%	Yes
Alexander	87.7%		Gaston	24.0%	Yes
Alleghany	88.2%		Gates	45.0%	Yes
Anson	44.3%	Yes	Graham	98.4%	
Ashe	86.1%		Granville	38.6%	Yes
Avery	68.2%		Greene	44.5%	Yes
Beaufort	48.1%	Yes	Guilford	49.3%	Yes
Bertie	34.3%	Yes	Halifax	44.9%	Yes
Bladen	47.1%	Yes	Harnett*	52.3%	Yes
Brunswick	77.1%		Haywood	72.5%	
Buncombe	62.1%		Henderson	67.2%	
Burke	75.9%		Hertford	40.3%	Yes
Cabarrus	61.9%		Hoke	36.5%	Yes
Caldwell	73.7%		Hyde	50.3%	
Camden	44.9%	Yes	Iredell	62.4%	
Carteret	56.5%		Jackson	85.0%	
Caswell	43.4%	Yes	Johnston	53.9%	
Catawba	78.2%		Jones	52.7%	
Chatham	61.7%		Lee	49.1%	Yes
Cherokee	80.0%		Lenoir	42.9%	Yes
Chowan	38.7%	Yes	Lincoln	82.6%	
Clay	92.4%		Macon	82.4%	
Cleveland*	51.4%	Yes	Madison	78.8%	
Columbus	54.6%		Martin	47.0%	Yes
Craven	37.1%	Yes	McDowell	71.2%	
Cumberland	27.5%	Yes	Mecklenburg	56.1%	
Currituck	54.3%		Mitchell	66.4%	
Dare	63.0%		Montgomery	70.2%	
Davidson	67.0%		Moore	55.5%	
Davie	77.2%		Nash	47.7%	Yes
Duplin	52.4%		New Hanover	55.1%	
Durham	53.9%		Northampton	45.0%	Yes
Edgecombe	40.4%	Yes	Onslow	23.6%	Yes
Forsyth	51.8%		Orange	58.7%	

Continued on the next page.

Table 8, continued

County	1964 Voter		County	1964 Voter	
	Turnout	Covered		Turnout	Covered
Pamlico	54.8%		Stokes	77.6%	
Pasquotank	43.6%	Yes	Surry	65.9%	
Pender	54.1%		Swain	89.3%	
Perquimans	46.2%	Yes	Transylvania	89.8%	
Person	48.7%	Yes	Tyrrell	55.7%	
Pitt	44.3%	Yes	Union	46.7%	Yes
Polk	89.0%		Vance	35.3%	Yes
Randolph	67.7%		Wake*	51.6%	Yes
Richmond	54.0%		Warren†	48.1%	
Robeson	40.0%	Yes	Washington*	50.5%	Yes
Rockingham*	50.4%	Yes	Watauga	85.5%	
Rowan	59.4%		Wayne	36.9%	Yes
Rutherford	65.3%		Wilkes	83.9%	
Sampson	62.9%		Wilson	38.7%	Yes
Scotland	39.2%	Yes	Yadkin	73.2%	
Stanly	70.5%		Yancey	77.6%	

Source: The measurement of 1964 presidential voter turnout was constructed using voter turnout data from Leip (2012). Estimates of the voting age population are from United States Commission on Civil Rights (1962). Coverage status under the 1965 VRA is from various volumes and sections of the *Federal Register*: United States (1965), United States (1966b), United States (1966a), United States (1966c).

Notes: The 1964 presidential voter turnout is calculated as a percentage of the voting age population. The counties labeled with a * are above the 50% threshold for coverage but were nevertheless covered (always-takers). The county labeled with a † is under the 50% threshold for coverage, but was not covered (never-taker). Note that Jackson county was covered by the 1970 extension of Section 5 of the VRA, while Wake county “bailed out” (was relieved) of its pre-clearance requirement in 1984.

Table B2: Non-Compliers by Measure of the Voting Age Population, Always-Takers (+) and Never-Takers (-)

County	VA	19+	20+	21+
Cleveland	+			+
Duplin		-		
Gaston			+	+
Guilford			+	+
Harnett	+			+
Hyde		-	-	-
Johnston		-		
Lee				+
Rockingham	+			+
Wake	+	+	+	+
Warren	-	-	-	-
Washington	+	+	+	+
Total non-compliers	6	6	6	10

Source: VA is the measure of the voting age population from United States Commission on Civil Rights (1962). The source does not confirm what age constitutes the “voting age population,” and instead simply refers to it as such. All of the specific age data (19+, 20+, 21+) come from the 1960 US Census as extracted from Minnesota Population Center (2011).

Notes: Counties with a minus sign are never-takers — counties that were assigned to coverage given the source’s measurement of the forcing (intent-to-treat) variable, but did not receive coverage. Counties with a plus sign are always-takers — counties that were not “assigned” to coverage given the source’s measurement of the forcing (intent-to-treat) variable, but which nevertheless received coverage. Counties that are coded correctly by the source are left blank or left out of the table. The voting age in North Carolina in 1964 was 21 years of age.

C Assessing Pre-Treatment Trends in Political Outcomes

In order to interpret the main results of the paper as causal, it's necessary to assume that there were no time-varying differences that may have affected *both* coverage status and post-1965 political outcomes. While this time-varying heterogeneity can be partially visualized in the graphical analysis presented in Appendix F (by observing whether the pre-1965 trends in political outcomes are parallel between covered and uncovered counties), and partially addressed with the inclusion of time-varying controls, this appendix more formally tests whether coverage status predicts changes in political outcomes prior to 1965. Although the parallel trends assumption cannot be directly tested, observing parallel pre-trends is important evidence in support of that assumption. Specifically, for county c and year t , I estimate

$$y_{ct} = \alpha_c + \gamma Time_t + \psi(Time_t \cdot covered_c) + \delta X_{ct} + \epsilon_{ct} \quad (1)$$

where y_{ct} is one of the political outcomes studied in this paper, α_c capture county fixed effects, T_t is a linear time trend, $covered_c$ is a dummy equal to one for covered counties, X_{ct} are a set of time-varying covariates that include population, the % of the population black, median household income and median schooling, and ϵ_{ct} is the idiosyncratic error. Equation 1 is estimated only for pre-1965, prior to Section 5 coverage.¹² The parameter ψ captures the differential linear trend by coverage status.

Table C1 presents the results from estimating equation 1. Each outcome from the main manuscript's analysis is presented in a separate specification. If trends in outcomes are the same between (eventually) covered counties and those left un-covered, we would expect the coefficient ψ to be statistically indistinguishable from zero. In all specifications except specifications 5 and 6 (where the dependent variable is Democratic vote share), the coefficient ψ is statistically insignificant.¹³ This evidence supports interpreting the estimated coefficients (with the exception of those for Democratic vote share) in the main analysis of the paper as causal.

Table C1 indicates that presidential and gubernatorial Democratic vote share was declining faster in covered counties relative to un-covered counties. A visual inspection of the trends in vote share in Appendix F, as well as restriction of the pre-1965 sample to presidential elections prior to 1964 indicate that the 1964 election plays a substantial role in these results. The 1964 presidential election pitted Lyndon B. Johnson against Barry Goldwater, and saw Johnson win in both an electoral college and popular vote landslide. North Carolina's 13 electoral college votes went to Johnson, the Democrat, but the remainder of the Deep South — from Louisiana to South Carolina (with the exception of Florida) — voted for Goldwater. This was a stark reversal in the party

¹²Note that the number of years of pre-1965 data differs by outcome.

¹³The estimate of ψ in specification (1) is just barely insignificant at conventional levels (p-value= ~ 0.13). Note however that the sign of the estimate indicates that covered counties were registering *more* blacks than covered counties. Insofar as we're concerned that the estimates presented in the paper *overestimate* the effect, the sign of this coefficient is helpful.

voting that had dominated the South since Reconstruction — Democrats, the party of Jim Crow and segregation, had previously enjoyed substantial support amongst Southern whites.

In the governor’s office, North Carolina Democrats were already moving away from a segregationist platform prior to the VRA (Beyle and Black, 1975). In much of the south, it was the Democratic primary where political competition could be found. But North Carolina had some of the most competitive general elections for state office in the decades before the VRA. Still, primary campaigns reveal something about changing racial attitudes in the state. The 1960 Democratic gubernatorial primary saw Terry Sanford win against I. Beverly Lake. The latter was an avowed segregationist, while Sanford spoke in moderate terms about “mixing in schools.” At least at the gubernatorial level, ardent segregationists were hard to find post-VRA. Some have claimed that this alone is evidence for the role of newly enfranchised blacks in politics, but such causal attribution is exceedingly hard to make (Beyle and Black, 1975).

Table C1: Results for Parallel Pre-Coverage Trends in Political Outcomes

	Black Voter Reg. (1)	White Voter Reg. (2)	W-B Gap (3)	Pres. Turnout (4)	Cong. Turnout (5)	Pres. Dem. Vote Share (6)	Gov. Dem. Vote Share (7)
Time ($\hat{\gamma}$)	0.16 (0.20)	0.077 (0.084)	-0.077 (0.19)	0.0035** (0.0017)	-0.020*** (0.0033)	-0.0023 (0.0025)	-0.0090*** (0.0032)
Time $\cdot$ covered ($\hat{\psi}$)	-0.068 (0.044)	-0.011 (0.024)	0.033 (0.056)	0.00021 (0.00099)	-0.00028 (0.0011)	-0.0081*** (0.0014)	-0.0072*** (0.0018)
Observations	200	200	200	400	700	400	400

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table above presents results from estimating equation 1 restricted to the sample of years prior to 1965. The number of sample years differs by outcome. All specifications include county fixed effects. All specifications include controls². An observation is a county-year pair.

On the eve of the VRA, while North Carolina as a whole had not fully shifted its party dominance in response to national realignment, the evidence from the pre-trends indicates that that shift was already taking shape. Covered counties were precisely those that were more likely to favor the Republican candidates over Democratic candidates than they had in previous presidential elections. By the 1968 presidential election, North Carolina voted heavily in favor of Richard Nixon, the Republican presidential candidate, though the Democrats lost less of the gubernatorial vote.¹⁴ Beyle and Black (1975) note that the Republican shift in North Carolina was fairly gradual with the exception of Nixon’s broadly popular campaigns.

As noted in the paper, this pre-VRA shift in Democratic vote share indicates that we should interpret the estimated coefficients on the model of vote share as partial correlations rather than causal effects. Realignment was a process rather than a distinct event, but it nevertheless appears

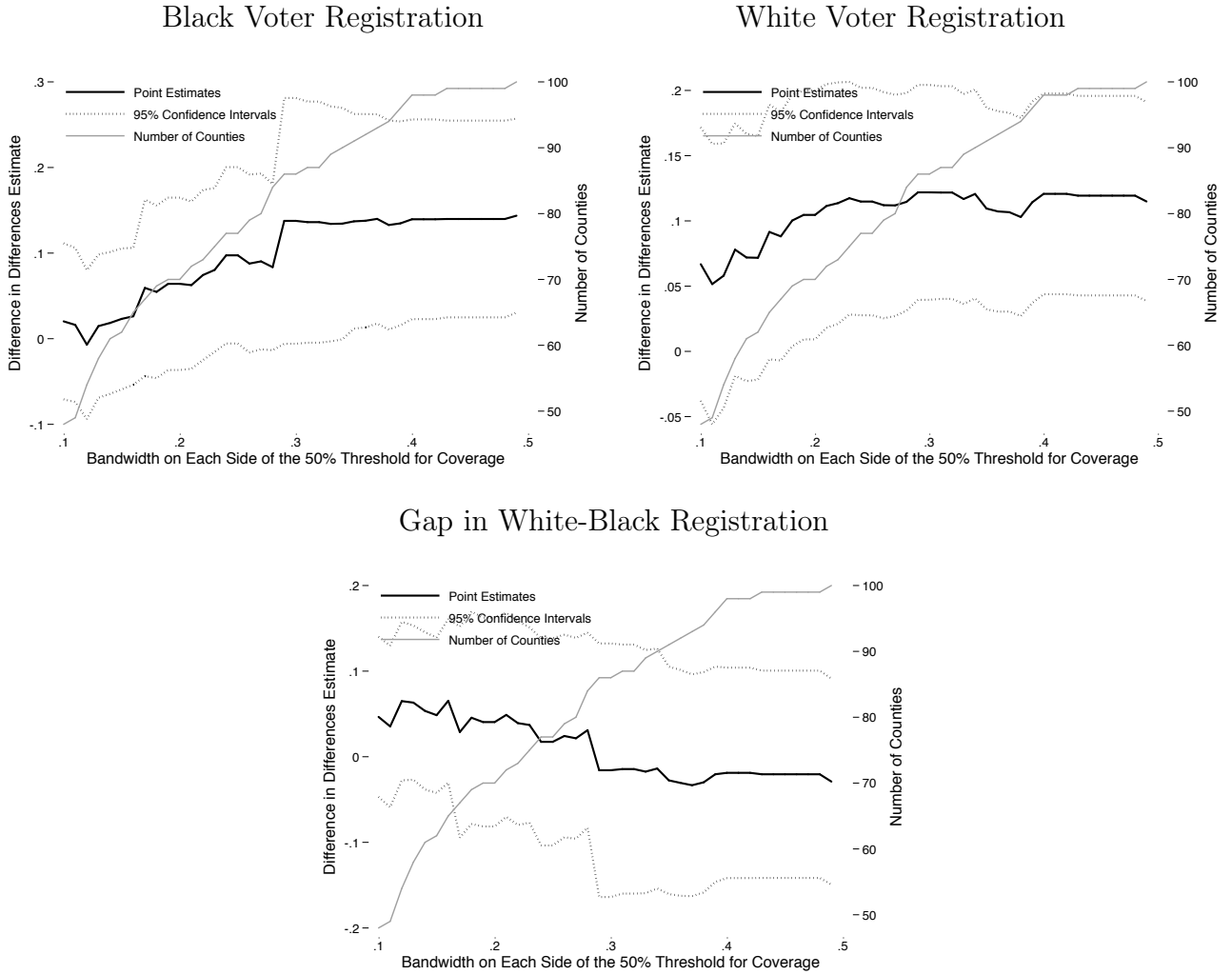
¹⁴North Carolina also followed with much of the South in heavily supporting the 1968 segregationist candidate, George Wallace.

that the realignment occurred differentially in covered versus uncovered counties. This might suggest some time-varying heterogeneity based on extant racial attitudes and political preferences — that is, that some counties were on different Democratic vote share trends as a consequence of some (relatively) fixed feature.

D Local Average Treatment Effects

In this appendix, I estimate the local average treatment effect (LATE) for those counties close to the threshold for coverage. In doing so, I assess the sensitivity of the main results of the paper and utilize information about how the Department of Justice Assigned counties to coverage.

Figure D1: Voter Registration by Bandwidth around the Threshold for Section 5 Coverage



Notes: The figure above presents plots of the estimated difference-in-differences coefficients and 95% confidence intervals from estimating equation 1 from the main body of the paper (including county and year fixed effects and using controls¹) for different sized bandwidths around the 50% threshold for Section 5 coverage. Recall that coverage was determined by voter turnout rates in the 1964 presidential election. Counties above 50% turnout were left un-covered, while those below 50% were covered. The plot on the left shows these results for black voter registration as a percent of the black voting age population. The plot on the right shows these results for the gap between white and black voter registration rates, each as a percent of the respective voting age population. In addition, each plot shows the number of counties that fall into a given bandwidth around the threshold for coverage. The x-axis in each plot is the bandwidth on either side of coverage. For example, the value 0.2 on the x-axis corresponds to the bandwidth from 30% voter turnout in the 1964 presidential election to 70% ($50\% \pm 20\%$). Seventy of North Carolina's 100 counties had 1964 voter turnout rates in this range.

Section 5 coverage in North Carolina was applied based on a threshold established on a continuous forcing (intent-to-treat) variable — voter turnout in the 1964 presidential election. Because of this, it's possible to estimate not only the average treatment effects described in the main results of the paper, but also local average effects that capture differences between political outcomes for counties

close to the threshold for treatment. These counties close to the threshold might be better argued to control for unobservable (time-varying) differences that might affect both coverage status and political outcomes, confounding the effects for the full sample of counties.¹⁵ In order to estimate these local average effects, I restrict the sample of counties to those counties in a (successively increasing) bandwidth around the 50% threshold for coverage (all integer bandwidths from ± 10 to ± 50 percentage points). I then estimate the difference-in-differences specification in the paper (equation 1 in the paper) on each of these restricted county samples. This specification includes both county and year fixed effects as well the time-varying controls¹ (see the paper).

Figures D1 and D2 present the results of estimating these restricted sample models for each of the political outcome variables of interest. I present the estimated coefficients visually since it allows for the most straightforward comparison of the coefficient magnitude and statistical significance for the many bandwidth sizes. On the x-axis of each of the individual plots is the bandwidth size around the 50% threshold for coverage, while the left y-axis presents the difference-in-difference estimate ($\hat{\beta}$). The right y-axis presents the number of counties that fall into a given bandwidth.

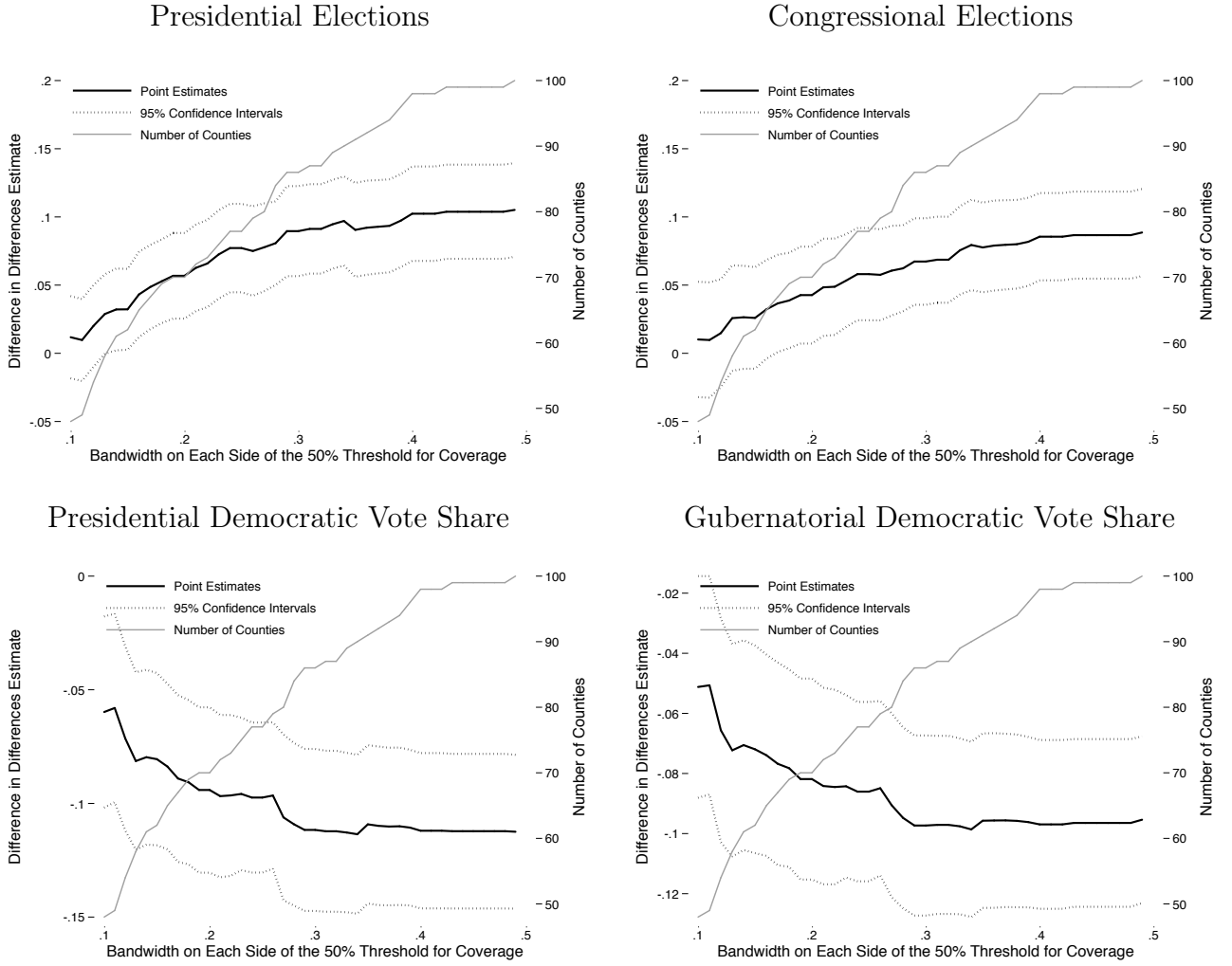
All of the plots indicate that the estimated effect of coverage is smaller in magnitude for small bandwidths. These results suggest that the local average effect of coverage for “moderately” discriminatory counties was smaller than the average effects presented in the main analysis of the paper. For example, counties within ± 20 percentage points on either side of the 50% threshold for coverage (counties with 1964 voter turnout between 30% and 70% of the voting age population), experienced an ~ 8 percentage point increase in their black voter registration (statistically significant at 90%, with 70 counties in the bandwidth), smaller than for the sample overall (with controls¹) of ~ 12 . The estimates for the smaller bandwidth are not statistically different from zero, although the sample size in those counties is quite small (only 50 counties fall within ± 10 percentage points of the threshold).

In the same ± 20 percentage point bandwidth, Figure D2 indicates that coverage caused a 7.5 percentage point increase in presidential voter turnout, a 5 percentage increase in congressional voter turnout, and a 10 percentage point drop in Democratic presidential vote share. This is a reduction approximately half in magnitude from the full-sample effects for turnout, and about a quarter for Democratic vote share. Despite this variation between restricted-sample and full sample effects, Figures D1 and D2 nevertheless indicate a good deal of stability both in the magnitude of the effect of coverage as well as its statistical significance.

How should we think about the LATE estimates relative to the estimates for the full sample? First, these results could be more “correct;” that is, better reflect the true causal effect of coverage. This would be the case if the restricted sample did indeed do a better job of accounting for time-varying unobserved heterogeneity that was missed in the full sample. However, it might also be the case that each result is in effect “true,” and that the differences lie in the fact that each of

¹⁵Though note that with the exception of Democratic vote share there is no evidence that trends in political outcomes were diverging between covered and uncovered counties prior to the VRA. See Online Appendix C.

Figure D2: Voter Turnout and Democratic Vote Share by Bandwidth around the Threshold for Section 5 Coverage



Notes: The figure above presents plots of the estimated difference-in-differences coefficients and 95% confidence intervals from estimating equation 1 from the main body of the paper (including county and year fixed effects and using controls¹) for different sized bandwidths around the 50% threshold for Section 5 coverage. Recall that coverage was determined by voter turnout rates in the 1964 presidential election. Counties above 50% turnout were left un-covered, while those below 50% were covered. The plot above-left shows these results for voter turnout in presidential elections as a percent of the voting age population. The plot above-right shows these results for voter turnout in congressional elections as a percent of the voting age population. The bottom left plot shows these results for vote share of the Democratic party in presidential elections, and the right plot Democratic party vote share in gubernatorial elections. In addition, each plot shows the number of counties that fall into a given bandwidth around the threshold for coverage. The x-axis in each plot is the bandwidth on either side of coverage. For example, the value 0.2 on the x-axis corresponds to the bandwidth from 30% voter turnout in the 1964 presidential election to 70% ($50\% \pm 20\%$). Seventy of North Carolina's 100 counties had 1964 voter turnout rates in this range.

these estimates are fundamentally different *quantities*. The true effects may be stronger or weaker for more or less initially discriminatory counties; that is, the effects may exhibit heterogeneity by initial levels of discriminatory practice captured by variation in the forcing variable. There are no differences in the extent to which the parallel trends test is violated for counties within this ± 10 sample around the threshold for coverage versus those at the extremes of the forcing variable (i.e. $< .4$ or $> .6$) — in both cases pre-VRA political trends do not predict coverage status.¹⁶

It is also possible that the LATE estimates are simply under-powered. The sample size relative to

¹⁶Results are available upon request. Also, see code in the associated replication files.

the variation in outcomes and coverage status is not particularly large (50 counties in total for the 10% window). Given many factors, including the measurement error in the registration data, it is should perhaps not be surprising that estimates for the small bandwidth are not distinguishable from zero.

Thus, the LATE results are suggestive of heterogeneity in the effects, but given the small sample it is not possible to know if these are true null effects or simply effects that are smaller for “moderately” discriminatory counties close to the threshold for coverage.

E Full Empirical Results

This appendix presents the full results for the models estimated and presented in the paper, including coefficient estimates for the control variables. Tables E1, E2 and E3 correspond to Tables 1, 2 and 3 in the paper, respectively.

In addition, I present results for the gap between white and black voter registration, results which are not presented in the main paper for reasons of space. These additional results are presented in Table E4. Examining the gap in registration helps to inform us about whether rates of white as opposed to black enfranchisement were different — that is, did registration rates for blacks outstrip those for whites. The evidence suggests only slightly, if at all. While the point estimates are all negative, the magnitude is small, and not statistically different from zero at conventional levels.

Table E1: Difference-in-Differences Results for Voter Registration

	Black Voter Registration			White Voter Registration		
	(1)	(2)	(3)	(4)	(5)	(6)
Post-1965 · covered ($\hat{\beta}$)	0.19*** (0.056)	0.15*** (0.057)	0.14** (0.054)	0.10** (0.040)	0.12*** (0.039)	0.13*** (0.039)
ln(population)		-0.13 (0.15)	-0.051 (0.23)		0.045 (0.086)	0.044 (0.12)
% population black		-2.30*** (0.37)	-3.46*** (0.80)		1.08*** (0.36)	0.95 (0.68)
Median income (0,000)		-0.0091 (0.0096)	-0.014 (0.035)		0.0073 (0.0047)	0.023* (0.013)
Median years schooling			-0.0087 (0.057)			-0.065** (0.028)
Observations	1695	1695	1195	1694	1694	1194

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table presents the estimates of β from estimating equation 1 in the main paper. The dependent variable in specifications 1-3 is black voter registration, while the dependent variable in specifications 4-6 is white voter registration. Each of these outcomes is measured as a percent of the relevant voting age population. All specifications include county and year fixed effects. An observation is a county-year pair. See Online Appendix A for more information on measurement and the sample size.

Table E2: Difference-in-Differences Results for Voter Turnout

	Presidential Elections			Congressional Elections		
	(1)	(2)	(3)	(4)	(5)	(6)
Post-1965 · covered ($\hat{\beta}$)	0.12*** (0.018)	0.11*** (0.018)	0.10*** (0.016)	0.10*** (0.017)	0.089*** (0.016)	0.088*** (0.016)
ln(population)		-0.15*** (0.055)	-0.17** (0.065)		-0.22*** (0.068)	-0.23*** (0.074)
% population black		-0.67*** (0.16)	-0.79*** (0.24)		-0.48* (0.26)	-0.46 (0.29)
Median income (0,000)		0.010*** (0.0033)	0.012 (0.0085)		0.022 (0.018)	0.022 (0.018)
Median years schooling			-0.027* (0.015)			0.0061 (0.026)
Observations	1200	1100	800	1100	1100	1100

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table presents the estimates of β from estimating equation 1 in the main paper. The dependent variable in specifications 1-3 is voter turnout in presidential elections, while the dependent variable in specifications 4-6 is voter turnout in congressional elections. Each of these outcomes is measured as a percent of the relevant voting age population. All specifications include county and year fixed effects. An observation is a county-year pair.

Table E3: Difference-in-Differences Results for Democratic Vote Share

	Presidential Vote Share			Gubernatorial Vote Share		
	(1)	(2)	(3)	(4)	(5)	(6)
Post-1965 · covered ($\hat{\beta}$)	-0.13*** (0.018)	-0.11*** (0.017)	-0.11*** (0.018)	-0.097*** (0.014)	-0.095*** (0.014)	-0.090*** (0.014)
ln(population)		0.035 (0.051)	0.093 (0.075)		-0.041 (0.047)	-0.061 (0.050)
% population black		0.52** (0.21)	0.47 (0.34)		0.51** (0.22)	0.40 (0.28)
Median income (0,000)		-0.0034 (0.0028)	-0.0030 (0.0092)		-0.0041 (0.0041)	0.0079 (0.0063)
Median years schooling			-0.012 (0.017)			-0.015 (0.016)
Observations	1200	1100	800	800	800	700

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table presents the estimates of β from estimating equation 1 in the main paper. The dependent variable in specifications 1-3 is two-party Democratic vote share in presidential elections; in specifications 4-6 it is for gubernatorial elections. All specifications include county and year fixed effects. An observation is a county-year pair.

Table E4: Difference-in-Differences Results for White-Black Voter Registration Gap

	W-B Registration Gap		
	(1)	(2)	(3)
Post-1965 · covered ($\hat{\beta}$)	-0.090 (0.065)	-0.034 (0.061)	-0.011 (0.059)
ln(population)		0.18 (0.16)	0.096 (0.25)
% population black		3.38*** (0.43)	4.40*** (0.93)
Median income (0,000)		0.016 (0.011)	0.037 (0.039)
Median years schooling			-0.057 (0.053)
Observations	1694	1694	1194

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

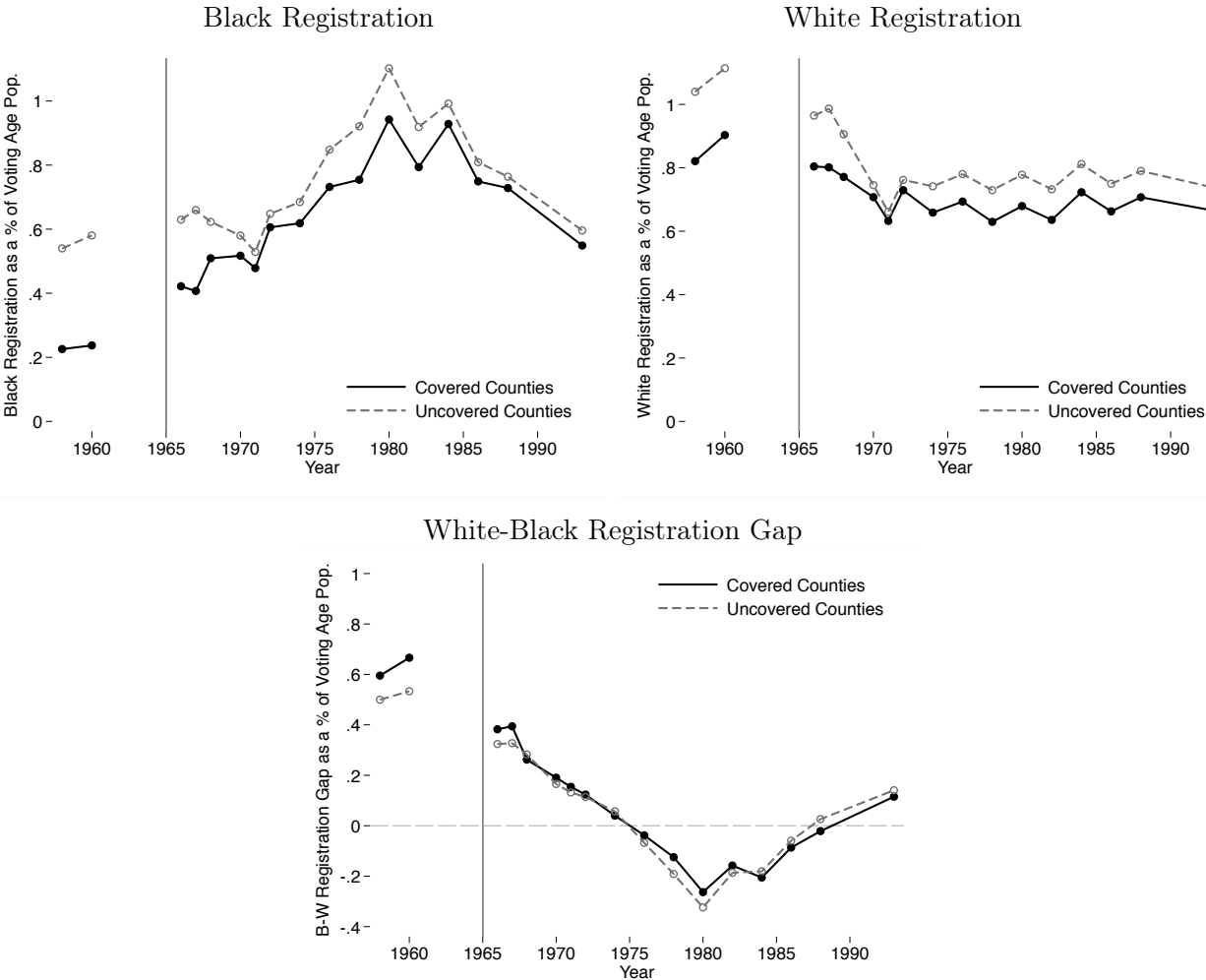
Notes: The table presents the estimates of β from estimating equation 1 in the main paper. The outcome is measured as the difference (white minus black) in registration rates each a percent of the relevant voting age population. All specifications include county and year fixed effects. An observation is a county-year pair.

F Graphical Analysis

This appendix presents the graphical results of the differences-in-differences analysis presented in the paper. Each plot in Figures F1 and F2 presents the average trends for a given outcome separately both before and after the application of Section 5 coverage, as well as separately for each covered and un-covered counties. The plots generally provide evidence supportive of the contention that trends in outcomes were parallel prior to the application of Section 5 coverage. The exception is Democratic vote in Figure F2.

Figure F1: Trends in Registration Rates By Coverage Status

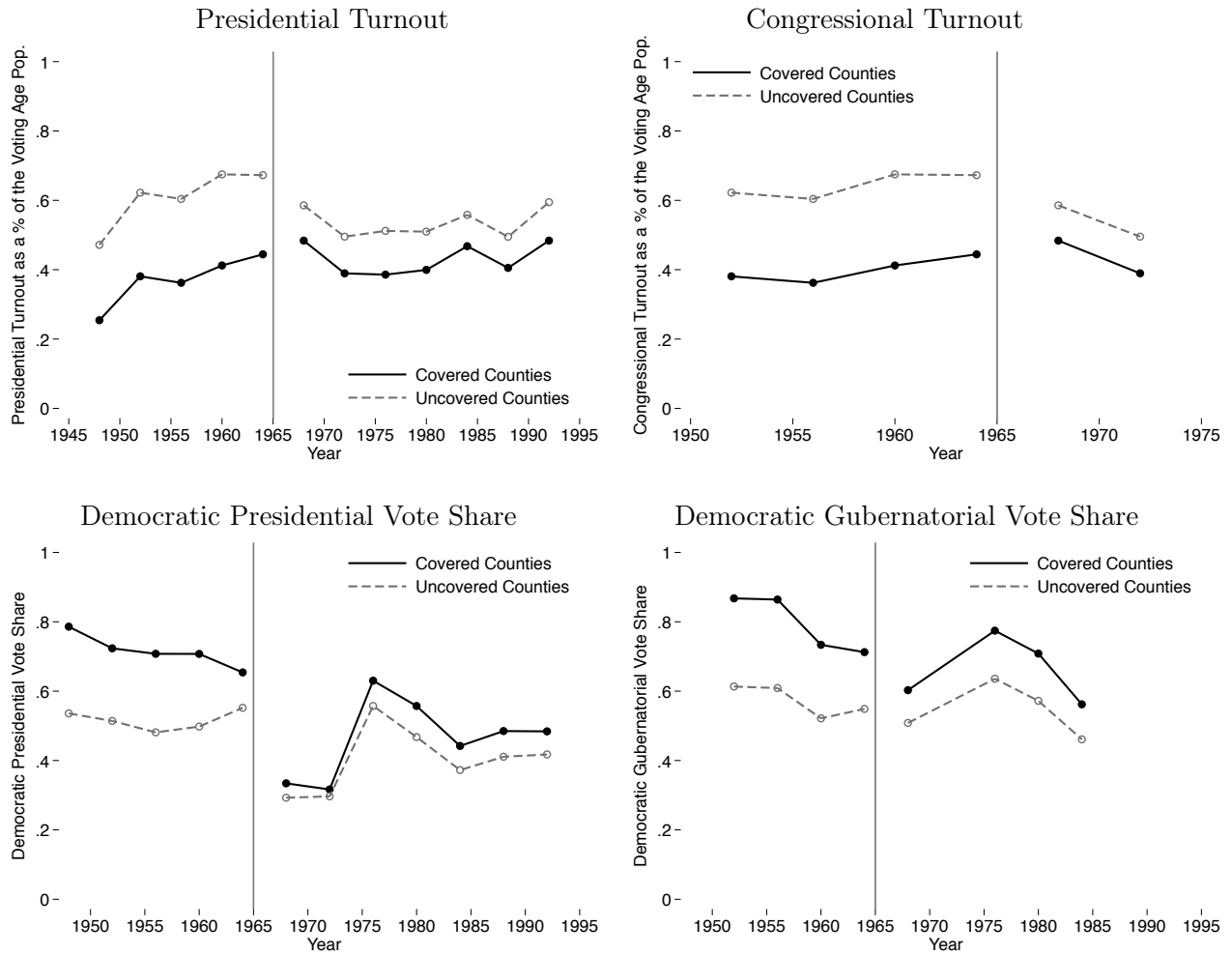
Black registration increased by more in covered counties post-1965 than un-covered counties. The white-black registration gap declined, but not by statistically significantly more in covered counties.



Notes: The plot above left presents trends in black voter registration as a percent of the relevant voting age population. The plot above right presents trends white voter registration rates, each as a percent of the relevant voting age population. The gap between white and black registration rates is the bottom plot. Trends are averages fit separately to North Carolina counties covered by the 1965 VRA (solid line), and those un-covered (dashed line). The vertical line at 1965 indicates the introduction of Section 5 coverage.

Figure F2: Trends in Turnout and Democratic Vote Share by Coverage Status

Voter turnout declined substantially in uncovered counties, but remained steady in covered counties.
Democratic vote share declined by more in covered than un-covered counties.



Notes: The plot above left presents trends in voter turnout in presidential elections as a percent of the relevant voting age population. The plot above right presents trends in voter turnout in congressional elections as a percent of the relevant voting age population. The plot below left presents trends in Democratic party vote share, and below right in gubernatorial Democratic vote share. Trends are averages fit separately to North Carolina counties covered by the 1965 VRA (solid line), and those un-covered (dashed line). The vertical line at 1965 indicates the introduction of Section 5 coverage.

G Generalizability of the North Carolina Results

This appendix presents a non-random sample of counties in each of the other covered states in the South with 1964 voter turnout and 1964 black voter registration similar to North Carolina counties close to the threshold for coverage (i.e. with 1964 voter turnout slightly less than 50%, and black voter registration around 32%, the average for North Carolina counties close to the threshold for coverage).¹⁷ One of the reasons that North Carolina was only partially covered by the 1965 VRA was that its discriminatory practices were less severe than many other southern states. As a result, we might wonder about how the effect sizes found in the paper’s main analysis generalize beyond the North Carolina context. Table G1 lists 1964 voter turnout and pre-VRA black voter registration for a sample of similar counties where we might expect these effects to hold.

Of course, southern states were not equal in the degree to which they disenfranchised blacks prior to the VRA, affecting the extent to which we can generalize this paper’s results beyond North Carolina.¹⁸ That North Carolina was partially covered while Mississippi was not, suggests that North Carolina’s pre-VRA disenfranchisement of blacks was less severe than was Mississippi’s. However, within-state variation in disenfranchisement still existed even in the most discriminatory of states, and many southern counties were similar to North Carolina counties close to the threshold for coverage.¹⁹ Finally, minority disenfranchisement in the South may have been pervasive and prominent, but covered states were hardly the only places that minorities faced disenfranchisement during the mid 20th century. Later extensions of the VRA’s coverage — to sub-state jurisdictions in California, Florida, Michigan and beyond — indicate that these results may tell us something about the effect of coverage at the local level outside of the South.

¹⁷Mississippi is excluded because no county had black voter registration above 22.7%.

¹⁸Just as the LATE estimates in Appendix D are a different quantity than the average North Carolina results, we might similarly consider the North Carolina results a local effect — that is, a different quantity — than the average results for the South as a whole.

¹⁹Table G1 in Online Appendix G presents examples of counties in which the effect of coverage may be very similar to the results found in North Carolina based on initial indicators of disenfranchisement. Furthermore, by interpreting the estimated effects found by this paper as average effects weighted by initial values of disenfranchisement and other covariates, the results can be generalized even further.

Table G1: Sub-State Jurisdictions in the Covered South Similar to North Carolina Counties Close to the Threshold for Coverage

County	1964 Voter Turnout	Pre-VRA Black Registration	County	1964 Voter Turnout	Pre-VRA Black Registration
<i>Alabama</i>			<i>South Carolina</i>		
Cleburne	45.2%	34.6%	Colleton	46.5%	30.3%
Lauderdale	49.6%	32.4%	Lancaster	46.3%	37.8%
Talladega	44.7%	32.1%	Union	45.5%	34.9%
<i>Georgia</i>			<i>Virginia</i>		
Cook	49.6%	34.2%	Buckingham	45.7%	37.4%
Spalding	46.7%	26.5%	Campbell	48.6%	43.4%
Turner	48.2%	30.2%	King William	45.4%	36.6%
<i>Louisiana</i>					
Avoyelles	48.5%	37.2%			
Natchitoches	45.3%	26.6%			
St. Helena	45.6%	26.9%			

Sources: Voter turnout is from Leip (2012), with voting age population from United States Commission on Civil Rights (1968). Voter registration is from United States Commission on Civil Rights (1968).

Notes: This is neither a complete nor random selection of counties that match on these two covariates. Close to the coverage threshold but still covered is 1964 voter turnout between 45% and 50% of the voting age population. Average black voter registration in those counties in North Carolina was 32%. Data for Mississippi on pre-VRA black voter registration by county is incomplete, and no county for which data is available had black voter registration above 22.7%, thus that state is excluded from this table.

H Results Using Alternative Registration Rate Denominators

This appendix re-estimates the main results of the paper using voter registration rates calculated using interpolated values of the relevant voting age population as opposed to using the percentages calculated by the sources (see Appendix A). These alternative calculations are only relevant for the registration variables, and only for the values pre-1970. As Table H1, the results using interpolated voting age population by race data are very similar to the main results presented in the paper, though smaller in magnitude and estimated with less statistical precision.²⁰

Table H1: Difference-in-Differences Results for Voter Registration Using Interpolated Voting Age Population

	Black Voter Registration			White Voter Registration		
	(1)	(2)	(3)	(4)	(5)	(6)
Post-1965 · covered ($\hat{\beta}$)	0.13** (0.050)	0.083* (0.046)	0.064 (0.043)	0.072* (0.037)	0.095*** (0.036)	0.10*** (0.036)
ln(population)		-0.21* (0.12)	-0.15 (0.20)		0.12* (0.064)	0.13 (0.083)
% population black		-2.16*** (0.36)	-3.50*** (0.85)		1.02*** (0.25)	0.79* (0.47)
Median income (0,000)		-0.0041 (0.0090)	-0.022 (0.033)		0.0064* (0.0037)	0.021* (0.011)
Median years schooling			-0.051 (0.061)			-0.061** (0.025)
Observations	1686	1686	1188	1694	1694	1194

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table presents the estimates of β from estimating equation 1 in the main paper. The dependent variable in specifications 1-3 is black voter registration, while the dependent variable in specifications 4-6 white voter registration. The outcome variables are calculated using interpolated values of the relevant voting age population to construct the denominator rather than using the percentages calculated by the sources described in Appendix A. All specifications include county and year fixed effects. An observation is a county-year pair.

²⁰This might be a consequence of measurement error in the denominator if the Civil Rights Commission or Voter Education Project who collected data on registration in the pre-1970 period had better access to the voting age population. It might also be a consequence of using in this appendix black voting age population paired with non-white registration totals where the contemporaneous sources utilized non-white voting age population. Finally, it is possible that there is systematic bias in the contemporaneous sources' measurements of the voting age population, though I do not have evidence of that.

Table H2: Difference-in-Differences Results for White-Black Voter Registration Gap

	(1)	(2)	(3)
Post-1965 · covered ($\hat{\beta}$)	-0.054 (0.059)	0.012 (0.050)	0.037 (0.047)
ln(population)		0.34** (0.14)	0.28 (0.21)
% population black		3.18*** (0.35)	4.28*** (0.85)
Median income (0,000)		0.011 (0.0097)	0.043 (0.035)
Median years schooling			-0.011 (0.057)
Observations	1686	1686	1188

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table presents the estimates of β from estimating equation 1 in the main paper. The dependent variable is the white-black voter registration gap. The outcome variables are calculated using interpolated values of the relevant voting age population to construct the denominator rather than using the percentages calculated by the sources described in Appendix A. All specifications include county and year fixed effects. An observation is a county-year pair.

I SUTVA Bias from Simulated Migration

One of the challenges to interpreting the main estimates presented in the paper as causal is that individuals may have moved between counties (or from/to North Carolina) as a consequence of coverage. If an individuals' decision to migrate is related to coverage, such migration constitutes interference between treatment and control units and violates the stable unit treatment value assumption (SUTVA), a key assumption necessary to estimate unbiased causal effects.

It is not possible with available migration data to directly estimate interference and thus bias.²¹ Therefore to better understand the bounds on this potential bias, this appendix presents a simulation of the main effects on black voter registration of the paper for different levels of migration.²² There are a number of different ways that migration can take place both within North Carolina and between North Carolina and other states. Migration that occurs within North Carolina — thus affecting both covered and uncovered counties simultaneously — has a greater potential to introduce bias into this paper's analysis than migration between North Carolina and other states.²³ Therefore, I simulate bias resulting from within-North Carolina migration only.

I further restrict the simulation to black migration and its potential bias to the estimates of the effect of the VRA on black voter registration. I assume that migration occurred as a consequence of the VRA because black migrants left uncovered counties and migrated to covered counties. I further assume that migrants *would have* registered in the uncovered county had they stayed, but chose to move to covered counties to take advantage of the legal protections afforded by Section 5 where they registered to vote.²⁴ Assuming that migrants would have registered in an uncovered county, while potentially less plausible than assuming that they would not have, is

²¹County level net migration-by-race data is available for much of the time period of study (Bowles et al., 2016; White, Mueser and Tierney, 1987; Guguit, Beale and Voss, 2010). However, net migration is a problematic measure of geographic spillovers. Consider the following example. If net migration in a given county in a given year is +100 people, it is not possible to know the underlying aggregate in and out migration — 100 people might have migrated in and 0 migrated out, or 10,100 might have migrated in and 10,000 migrated out. If those who migrated in did so to register, and those who migrated out did so because they weren't going to register, then the difference between in and out migrants (i.e. net migration) tells us very little about how we should expect registration rates to change. In an (implausibly) extreme example, one can imagine a scenario in which all black registrants in a given covered county might be migrants from an uncovered county who moved to take advantage of coverage, replacing previous black residents who would not have registered and instead migrated away. Net migration might therefore be zero in the covered county, even though the black population was wholly replaced. Even on a small scale this type of migration introduces bias, but cannot be measured with net migration rates. This ambiguity in the extent to which net migration measures the theoretical construct of interest is in addition to the question of whether net migration should be thought of as an outcome in its own right. Thus, a statistical model in which migration is simply *controlled for* might introduce post-treatment bias.

²²The simulation focuses on the results for registration rates alone since a registration simulation requires fewer assumptions about behavior by race, patterns of migration and individual political behavior as compared to voter turnout and vote share. I further focus on black registration rates because it's not clear ex-ante what white migration patterns we should expect as a consequence of coverage. Poor whites may have migrated to covered counties to register, but it may also have been the case that whites migrated out of covered counties to avoid having their votes diluted by a newly enfranchised minority.

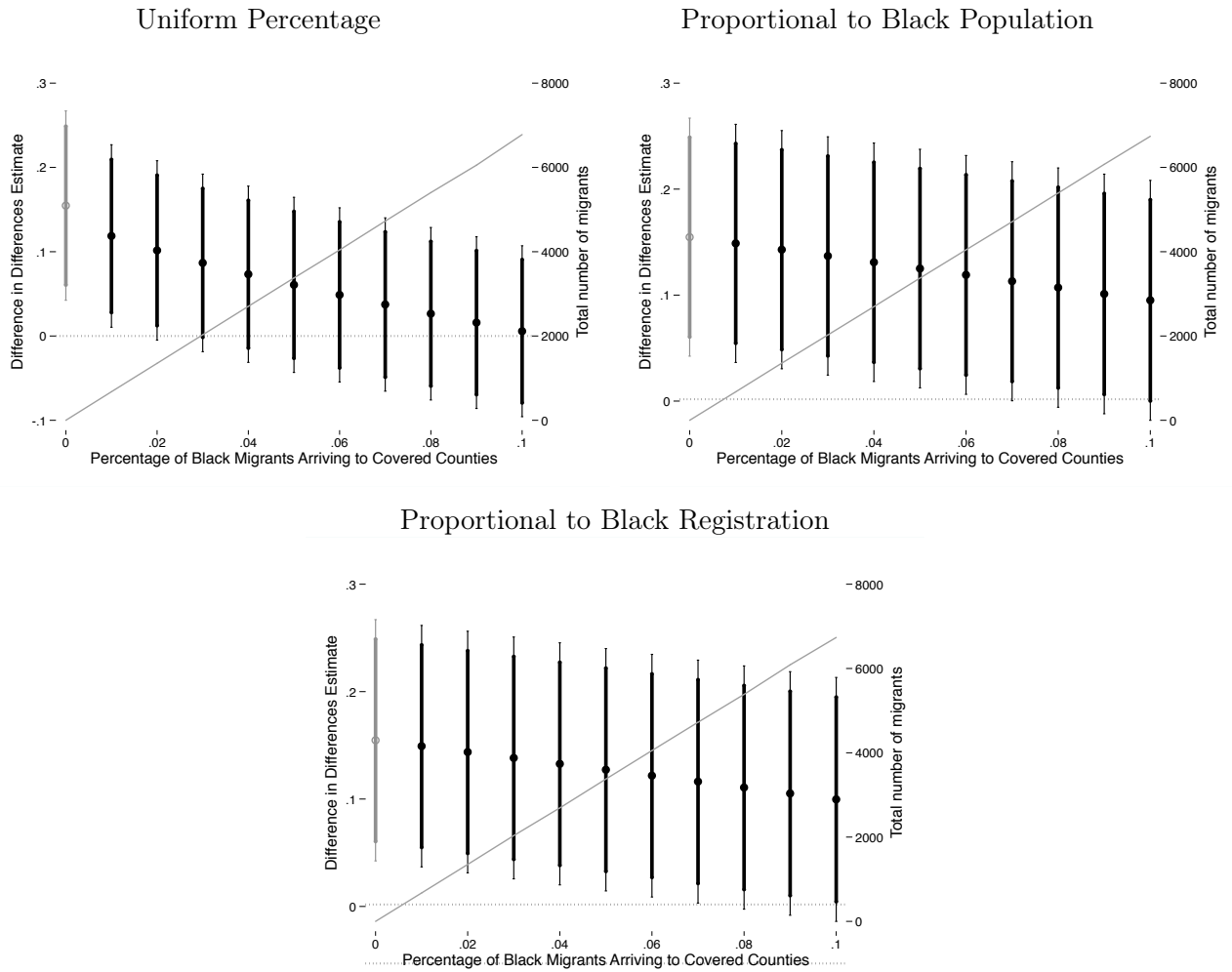
²³Within-North Carolina migration provides an upper bound on SUTVA bias.

²⁴Note that migration in the opposite direction — migration of *non-registrants* from covered counties to uncovered counties is entirely analogous. It's possible to think of the simulated migration as comprising either type of migration pattern or some mix of the two.

the more extreme form of potential bias since it reduces registrants in uncovered counties and increases them in covered counties. This assumption allows us to think of the simulated estimates as providing an upper bound on the bias that would have occurred for a given migration rate.

Because bias-inducing migration occurs from migration from uncovered to covered counties, I simulate *re-moving* registrants from covered counties and moving them (back) to uncovered counties. I assume a uniform percentage of registrants in covered counties migrated from uncovered counties, and then vary that percentage in the simulation.²⁵ I simulate percentages of migrants per county to ensure that I do not simulate negative voter registration in any given county.

Figure I1: Simulation of Black Voter Registration Responses to Black Migration



Notes: The above plots present simulations that re-estimate equation 1 from the paper using controls1, county as well as year fixed effects. Each point estimate is a re-estimate of equation 1 with 90 and 95% confidence bounds. The plots also indicate the average total migrants that I simulate moving (by year) for a given uniform arrival percentage. The plot above assumes that migrants left uncovered counties in uniform proportion and arrived to covered counties in uniform proportion. The plot above right assumes migrants left uncovered counties in proportion to their black population, but arrived in covered counties in uniform proportion. Finally the lower plot assumes migrants left uncovered counties in proportion to their black registration and arrived in covered counties in uniform proportion.

²⁵Note that I also adjust relevant covariates and relevant denominators to account for the simulated migration. The percentage of migrants in covered counties is assumed to be uniform across counties and across years. But because the total number of registrants varies by county and by year, the total number of assumed/simulated migrants varies by year.

In Figure I1 below, I present plots of estimates of β from equation 1 in the paper for different assumed percentages of black registrants that are migrants in covered counties (i.e. in which I *simulate* removal of those migrants “back” to uncovered counties). In all simulations I assume a uniform percentage in covered counties, but employ different assumptions about the patterns of out migration from uncovered counties. The first is uniformity — uncovered counties sent a uniform percentage of their black registrants to covered counties. The second is proportional to black population — counties sent migrants in proportion to their black population (i.e. as a percentage of total uncovered county black population). And the final is proportional to black registration — counties sent migrants in proportion to their registration (i.e. as a percentage of total uncovered county black voter registration).²⁶ In all cases I assume that migrants arrived in uniform percentage in covered counties.²⁷ None of these assumptions are without flaws, but lacking knowledge on actual migration patterns, they at the very least allow us to assess the sensitivity of a given assumption.²⁸

For the uniform simulation in Figure I1, the estimate of β is no longer significant at the 95% level given 2% migration, and no longer significant at the 90% level given 3% migration — i.e. if 3% of a covered county’s registrants were migrants from uncovered counties who would have registered in the uncovered county, the causal effect of coverage would no longer be statistically different from zero at the 90% level. The percentages for the population proportion are 7% and 10%, and for registration proportion the same. Migration of 3% represents approximately 2,000 total registrants in a given year, while 10% migration represents approximately 7,000 total registrants.

How do these simulations correspond with known migration data? To the author’s knowledge, there are no systematic sources of total in and out migration by race for the period under study in North Carolina. The census asks where individuals lived 5 years ago (same/different house, same/different county, same/different state). Although not differentiated by race or *which* county individuals previously lived in, on average in 1970 11000 people lived in a different county in North Carolina in 1965.²⁹ This is more in line with the 10% migration percentage than the 3%, suggesting that if migration followed a uniform pattern (and adhered to the many other assumptions of the simulation) that the true causal effect of coverage might be statistically indistinguishable from

²⁶An example of the procedure follows. For 2% migration in the uniform percentage plot, I assume that 2% of black registrants in each covered county were migrants from uncovered counties. This amounts to 1351 total migrants in covered counties. I simulate moving these individuals back to uncovered counties (and registering them to vote). I assume that they arrived uniformly from uncovered counties — that is, each uncovered county sent 22 of its registrants to covered counties. In the proportional assumptions, I assume that this number being sent to covered counties varies depending on black population and black registration amongst uncovered counties.

²⁷Again, this assumption ensures that when I simulate removing them that I don’t generate negative voter registration.

²⁸An alternative to these three distinct assumptions would be a more explicit model of migration patterns taking into account geography. Such models often assume that migration occurred between neighboring pairs of treatment and control units, or was more likely within specific geographic bounds. While potentially useful in some settings, it’s not clear that the benefits of such geographic specificity of modeled migration patterns offsets the cost of their implementation. There are limited studies of actual interference bias in settings such as this, which leaves us with limited theoretical priors about the specific pattern of migration.

²⁹The average migration rates are not substantially different by coverage status — approximately 10300 in uncovered counties and 12800 in covered counties.

zero. However, given the myriad of assumptions and the measurement issues in the migration data, it is not possible to say for certain.

Finally, I note the following. While migration is a threat to the interpretation of the estimates in the paper as causal effects, from an historical and policy standpoint, general equilibrium effects such as migration are interesting outcomes in their own right. To the extent possible, further research should seek out better data to assess the extent to which individuals responded to coverage under the VRA by moving between covered and uncovered counties.

J Short, Medium and Long-Run Results

In this appendix, I assess the short, medium and long-run effects of the Voting Rights Act. The estimated effect presented in the paper an average effect over the entire post-VRA period (approximately twenty five years), and there is ample theoretical interest in whether that average is due to immediate changes in the first few years after the act, or whether average changes over the entire period were driven by medium and longer-run dynamics.

To examine these questions of timing and duration, I use OLS to estimate

$$y_{ct} = \alpha_c + \gamma_t + \beta_1(\mathbb{1}_t^{1965-1970} \cdot cov_c) + \beta_2(\mathbb{1}_t^{1970-1982} \cdot cov_c) + \beta_3(\mathbb{1}_t^{post-1982} \cdot cov_c) + \delta \mathbf{X}_{ct} + \epsilon_{ct} \quad (2)$$

where the post period is broken into the short-run (1965-1970), the medium run (1970-1982), and the long(er) run (post-1982). I include controls¹ in the specification which allow for the maximal number of years (as opposed to controls², which are not available for all years for which there is outcome data).

The breakpoints are chosen ex-ante rather than inductively determined, but correspond to important historical progress of the VRA. When the VRA was first signed into law in 1965, it was given a five year expiration. Thus, the first breakpoint occurs five years after the VRA was first signed. The law was renewed (and expanded in 1970) with the Department of Justice noting that while minority political participation had improved, many jurisdictions were nevertheless ignoring the pre-clearance requirement. The first five years were also a period of substantial uncertainty about the nature of the law. It wasn't until 1969 and *Allen v. State Board of Education* that the scope of the law with regards to voting related practices was established.³⁰ After the law was first renewed, it went through another series of 5 year renewals. Each subsequent renewal became more likely than the last. Finally in 1982 Congress renewed the act (and specifically Section 5) for a more substantial period of time, 25 years.³¹

Table J1 presents the results from estimating equation 2 for each of the political outcomes studied in this paper. The estimates are in reference to the pre-VRA period. Thus, the estimate of β_2 , for example, is the difference in outcome between covered and uncovered counties from pre-1965 to the period 1970-82. In the case of specification 1 and black voter registration, the effect size grows over time, from 14 percentage points to 25 percentage points. White voter registration also grows from the short to medium run, but then levels off. While never statistically distinguishable from zero, the gap between white and black registration rates, is largest during the most recent period. Finally, turnout and vote share percentages are generally stable over time.

These results suggest that the impact of the VRA grew overtime with respect to registration. This was potentially a consequence of the growth in Department of Justice reviews of proposed voting changes and legal challenges to existing discriminatory laws. As these discriminatory laws were

³⁰And not until 1971 was it established that polling place location fell under Section 5 in *Perkins v. Matthews*.

³¹There are not congressional turnout estimates post-1982 in this dataset.

abolished, black registration became increasingly common.

Table J1: Short, Medium and Long-Run Effects of Section 5 on Political Outcomes

	Black Voter Reg. (1)	White Voter Reg. (2)	W-B Gap (3)	Pres. Turnout (4)	Cong. Turnout (5)	Pres. Dem. Vote Share (6)	Gov. Dem. Vote Share (7)
1965-1970 · covered ($\hat{\beta}_1$)	0.11** (0.052)	0.084** (0.041)	-0.028 (0.058)	0.12*** (0.017)	0.086*** (0.016)	-0.13*** (0.018)	-0.12*** (0.018)
1970-1982 · covered ($\hat{\beta}_2$)	0.16** (0.060)	0.14*** (0.040)	-0.017 (0.063)	0.095*** (0.017)	0.098*** (0.023)	-0.11*** (0.018)	-0.076*** (0.015)
Post-1982 · covered ($\hat{\beta}_3$)	0.21*** (0.070)	0.14*** (0.040)	-0.074 (0.073)	0.11*** (0.021)		-0.11*** (0.018)	-0.11*** (0.018)
Observations	1695	1694	1694	1100	1100	1100	800
Controls1	✓	✓	✓	✓	✓	✓	✓

Standard errors robust to clustering at the county level presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table above presents results from estimating equation 2. The number of sample years differs by outcome. All specifications include county and year fixed effects. All specifications include controls1. An observation is a county-year pair. There are not congressional turnout estimates post-1982 in this dataset.

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